GOM25: A Stable Geodetic Reference Frame for Assessing Infrastructure Health and Coastal Hazards in the Gulf of Mexico Region

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Abstract: The Gulf of Mexico (GOM) region faces acute challenges from coastal subsidence, sea-level rise, and fault-related ground deformation, necessitating a geodetic reference frame capable of isolating minor regional- and local-scale ground deformation from the background signal of rigid plate rotation and translation. We present the Gulf of Mexico Reference Frame 2025 (GOM25), a temporally stable and spatially consistent datum aligned with the International Global Navigation Satellite Systems (GNSS) Service Reference Frame 2020 (IGS20). GOM25 is realized through a densified network of 150 continuous GNSS stations, located in the northern GOM region and anchored to the stable interior of the North American Plate. The frame achieves a stability of 0.3 mm/year horizontally and 0.5 mm/year vertically, providing a reliable reference foundation for the greater Gulf region (15°–37° N; 110°–75° W). To ensure practical accessibility, we provide a one-set seven-parameter transformation for Earth-centered, Earth-fixed (ECEF)-XYZ coordinates, from which we derive the corresponding transformations for displacements and site velocities between IGS20 and GOM25. Applying this regional frame yields quantitative insights relevant to infrastructure and hazard management; two examples exemplify its utility: (1) a mean sea-level rise rate of 2.7 ± 0.4 mm/year during the last half century along the GOM coast and South Atlantic Bight, and (2) an average uplift rate of 1.3 ± 0.6 mm/year in the southern High Plains over the last two decades—both measured relative to GOM25, physically the stable Gulf coastal plain. GOM25 provides a critical foundation for reconciling sea-level measurements, constraining postglacial rebound models, and monitoring the structural health of onshore and offshore infrastructures, supporting interdisciplinary efforts in coastal flooding and inundation mapping, wetland loss mitigation, urban subsidence assessment, and resilient planning across the region. **DOI: 10.1061/JSUED2.SUENG-1721**. *© 2026 American Society of Civil Engineers.*

Practical Applications: The GOM region is home to major metropolitan areas, thousands of kilometers of pipelines, hundreds of ports, extensive levee systems, and numerous offshore platforms. To precisely monitor long-term regional- to local-scale ground movement and structural deformation, engineers need a stable reference frame that removes the broad plate motion signal inherent in GNSS measurements. This paper presents GOM25, a new stable regional reference frame for the Gulf. It is built from 150 continuous GNSS stations anchored to stable parts of the North American Plate and achieves a frame stability of 0.3 mm/year horizontally and 0.5 mm/year vertically. Practitioners can easily apply GOM25 using a single set of seven transformation parameters, which we provide along with ready-to-use Python scripts. These tools convert coordinates, displacement time series, and velocities between the International GNSS Service Reference Frame 2020 and GOM25 without the need for daily transformation parameters. Engineers and planners can use GOM25 to assess long-term infrastructure health (e.g., pipelines, platforms, and levees), calibrate Interferometric Synthetic Aperture Radar (InSAR) or uncrewed aerial vehicle (UAV) surveys, map subsidence and flood risk, and support resilient coastal design.

Introduction

The Gulf of Mexico (GOM) lies between the North American and Caribbean plates and is bordered by the US to the north, Mexico to the west and south, and Cuba to the southeast. Its coastline exceeds 5,500 km (mainland only) and supports major population centers and extensive energy, transportation, and industrial infrastructure across five US states (Florida, Alabama, Mississippi, Louisiana, and Texas) and five Mexican states (Tamaulipas, Veracruz, Tabasco, Campeche, and Yucatán). The region is a hub of critical onshore and offshore infrastructure, including seawalls, levees, and a dense network of pipelines, ports, refineries, and production platforms, all of which require a temporally stable and spatially consistent geodetic reference frame to assess their vulnerability to natural hazards and ensure their long-term resilience.

Although the GOM region is often treated as tectonically stable at the plate scale, ground deformation is spatially variable and locally consequential for coastal communities. In the Mississippi Delta, subsidence amplifies erosion and flood risk; along the Texas Gulf Coast, severe subsidence associated with fluid extraction occurs in addition to fault-related deformation. The impacts of coastal subsidence are compounded by sea-level rise, resulting in increased coastal flooding (e.g., Miller and Shirzaei 2021; Bilskie et al. 2022; Wang et al. 2024; Ohenhen et al. 2024). Comparable processes in Mexico and Cuba, driven by faulting, subsidence, and sea-level rise, significantly impact urban and coastal areas, intensifying flooding, erosion, and infrastructure vulnerabilities.

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The Global Navigation Satellite System (GNSS) has become a primary geodetic technique for continuous, high-precision monitoring of ground deformation. It also provides essential ground truth for calibrating and validating other geodetic sensors, including interferometric synthetic aperture radar (InSAR), airborne and terrestrial light detection and ranging (LiDAR), and uncrewed aerial vehicle (UAV) photogrammetry. Typically, initial positions and velocities derived from GNSS observations are expressed in a global reference frame—currently International GNSS Service (IGS) Reference Frame 2020 (IGS20) (Rebischung et al. 2022), which is the GNSS-specific realization of the International Terrestrial Reference Frame 2020 (ITRF2020) (Altamimi et al. 2023). IGS20 is defined by imposing no-net-translation (NNT) and no-net-rotation (NNR) conditions to a global network of over 330 long-history GNSS stations, ensuring its origin aligns with the geocenter and its scale and orientation are consistent with ITRF2020. For this study, we adopt IGS20 because it is the natural realization for GNSS data processing (consistent with IGS products), and its differences from ITRF2020 are at the submillimeter level—negligible for regional deformation analyses.

Although alignment with global reference frames ensures international consistency, site velocities within these frames primarily reflect secular plate motion and spatially correlated geophysical signals rather than localized stability. Large-scale signals from rigid-body motion—comprising both plate rotation and translational shifts—as well as glacial isostatic adjustment (GIA) often mask subtle deformations. Consequently, environmental, engineering, and geophysical applications focused on the regional or local scale require a stable regional reference frame to isolate site-specific or structural movements that would otherwise be obscured by these global kinematic trends.

In 2014, our research team established an initial regional reference frame, the Stable Gulf of Mexico Reference Frame, utilizing approximately 10 years (2005–2014) of continuous GNSS observations from 13 stations distributed within the GOM region (Yu and Wang 2016, 2017). This preliminary frame was aligned with the IGS08 global frame. Building upon this foundation, we subsequently developed GOM20 in 2020 to align with the IGS reference frame at that time, IGS14 (Wang et al. 2020). GOM20 and its predecessors have effectively supported numerous regional deformation studies, including development of regional crustal-movement velocity model (e.g., Snay et al. 2025), regional subsidence analysis (e.g., Ellis et al. 2023), fault activity monitoring (e.g., Liu et al. 2019), delineating natural subsidence (Zhou et al. 2021), and sea-level rise assessments (e.g., Liu et al. 2020), and offshore structural (platform) health monitoring (Wang 2023a).

Accurate characterization of the aforementioned deformation processes—many with strong anthropogenic drivers—requires continuous, precise, and long-term geodetic observations referenced to a stable reference frame. To meet this need, we present the Gulf of Mexico Reference Frame 2025 (GOM25), a temporally stable and spatially consistent datum for the GOM region. Several recent developments necessitated this update from its predecessor, GOM20.

First, the global geodetic community has transitioned to the IGS20 reference frame, which features updated station positions and velocities, refined Earth orientation parameters, and enhanced geophysical models. Aligning our regional frame to this most recent global standard is essential to maintain consistency and accuracy with contemporary geodetic products. Second, by 2025, an additional 5 years of GNSS data and a substantially expanded network of reference stations had become available compared with those used for GOM20, enabling improvements in both temporal coverage and spatial station density across the Gulf region.

Beyond the frame itself, we provide generalized methods and open-source Python tools for converting positional coordinates, displacement time series, and site velocities between global and regional reference frames, along with a detailed procedure to assess the spatial scope of this regional frame's applications.

In the following sections, we describe the data sources, reference station selection criteria, and methodology for realizing the regional reference frame and transforming coordinates, displacements, and velocities. We also detail the assessment of the frame's stability and its applicable geographic scope. Finally, we highlight application examples that demonstrate how GOM25 supports high-precision studies of regional-scale uplift, subsidence, and sea-level change in the GOM region.

Data and Methods

GNSS Stations

A dense network of continuously operating GNSS stations has spanned the greater GOM region since around the 2010s, providing daily high-precision positioning data for ground deformation studies. As illustrated in Fig. 1, approximately 1,000 continuous GNSS stations each meet two selection criteria: they provide more than 1,000 daily observational files, and their data span at least 4 years, acknowledging that many stations contain gaps. A substantial subset has data spanning over a decade. These stations are operated by diverse agencies and networks, including University NAVSTAR Consortium (UNAVCO), the Texas Department of Transportation (TxDOT), SmartNet North America, the US Coast Guard, the Center for GeoInformatics (C4G) at the Louisiana State University, the Harris-Galveston Subsidence District, HoustonNet at the University of Houston, trans-boundary, land, and atmosphere long-term observational and collaborative network (TLALOCNet) in Mexico, and continuously operating Caribbean GPS observational network (COCONet) in the Caribbean. GNSS raw data, specifically from receiver independent exchange (RINEX) format files, are openly accessible through data centers such as UNAVCO, the National Geodetic Survey (NGS), the Crustal Dynamics Data Information System (CDDIS), and local archives.

Most GNSS antennas in the region are mounted on low-rise building–based monuments, where a steel pole affixes the antenna to a building's roof or sidewall. In general, the building itself has a foundation extending several meters into more stable subsurface layers, which helps mitigate short-term soil-related deformations such as shrink–swell cycles during droughts (Yang et al. 2016; Welch et al. 2024). This configuration enhances the reliability of the resulting GNSS time series for detecting long-term trends in ground deformation studies.

Positioning Method

GNSS antenna-receiver units do not directly yield highly accurate positions; instead, they record the distances between satellites and the ground antenna at discrete time intervals—typically every 15 s for static positioning. Precise point positioning (PPP) is an absolute positioning method used to derive positions from raw GNSS data (Zumberge et al. 1997). PPP has become popular for geological hazards and structural health monitoring because it eliminates the need for local base stations, delivers consistent accuracy, and facilitates seamless integration of GNSS-derived site velocities from different studies (e.g., Bao et al. 2018; Wang 2013). When processed with precise orbit and clock products and transformed to a specified reference frame using established transformation parameters, all solutions are consistently aligned to the same global reference frame.

In this study, we employed Earth-centered, Earth-fixed (ECEF) XYZ coordinate time series from the Nevada Geodetic Laboratory (NGL) at the University of Nevada (Blewitt et al. 2018). NGL

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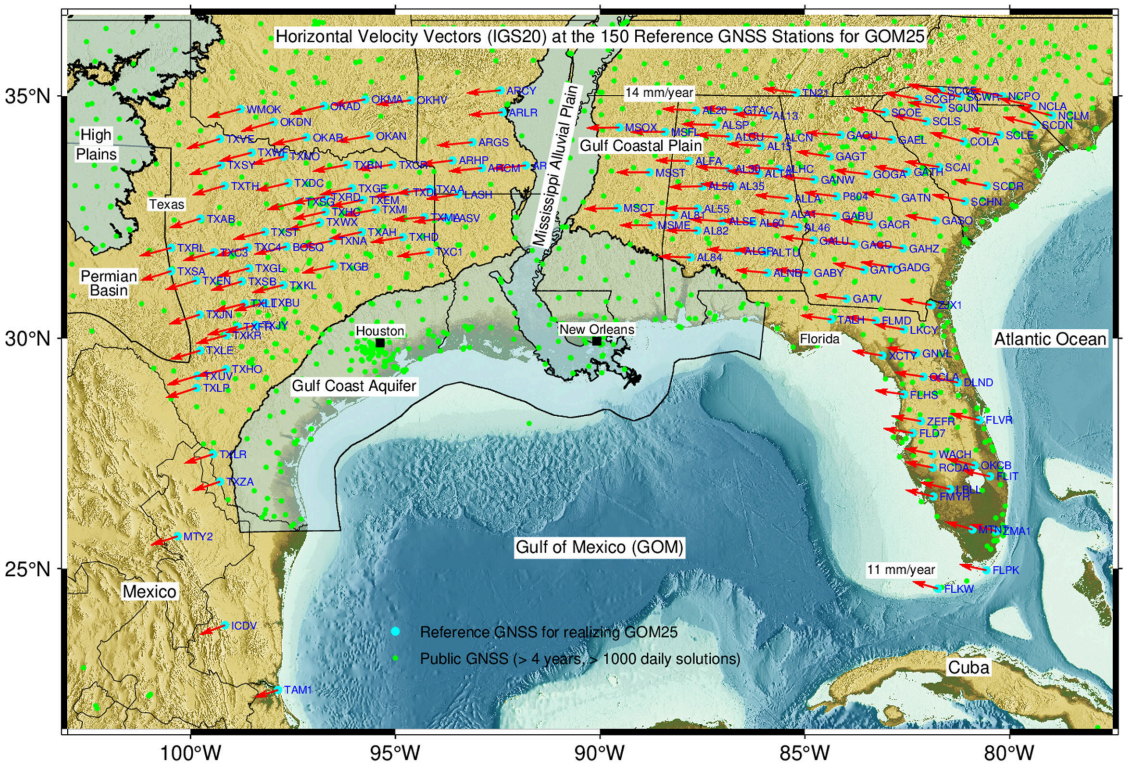


Fig. 1. Distribution of 150 reference GNSS stations used to realize the stable Gulf of Mexico Reference Frame 2025 (GOM25) and their horizontal velocity vectors with respect to the International GNSS Service Reference Frame 2020 (IGS20). Dots show public GNSS stations (>4 years and >1,000 daily solutions) in the GOM region. The map highlights three major deformation zones excluded from the reference station selection: the Permian Basin (Bureau of Economic Geology 2025) and High Plains (Qi 2010), the Gulf Coast Aquifer System (Casarez 2020), and the Mississippi Alluvial Plain (Painter and Westerman 2018). [Topo-bathymetric data from GEOBCO Compilation Group 2024; base map generated using The Generic Mapping Tools, v 6.3.0 (Wessel et al. 2019).]

uses PPP within the GipsyX (version 2.3) software package to generate single-receiver, phase-ambiguity-fixed daily solutions that are initially aligned to the satellite orbit reference frame. These daily solutions are then transformed to the IGS20 reference frame using the time-dependent Helmert parameters provided by Jet Propulsion Laboratory (JPL), specifically X-files (Bertiger et al. 2020).

The NGL processing uses only GPS observations to derive daily position solutions, even though some stations also record signals from other GNSS constellations (e.g., Galileo, GLONASS, and BeiDou). Detailed processing strategies can be found in NGL's technical documentation at its website (Blewitt et al. 2018). Additionally, for stations in the greater Houston area, we performed local processing at the University of Houston Geodesy Laboratory (HoustonNet) under the same PPP protocols, ensuring consistency in the derived time series (Wang et al. 2022).

To study ground deformation, we first transform the ECEF-XYZ coordinates from the IGS20 frame to the GOM25 frame using the one-set seven-parameter transformation. The resulting GOM25 ECEF-XYZ coordinates are then converted to geographic coordinates (longitude, latitude, and ellipsoidal height) referenced to the GRS80 ellipsoid. For each station, we define a local topocentric coordinate system [east, north, up (ENU)] where the east and north axes lie in the local horizontal plane. This is accomplished by rotating the ECEF-XYZ coordinate differences (relative to a stable reference epoch) into the ENU frame using the station's geographic position. The east and north components track horizontal surface deformation, and the up component (ellipsoidal height change) tracks vertical land motion.

In theory, vertical deformation should be measured using orthometric heights; however, for crustal deformation studies, ellipsoidal height changes are consistent with orthometric height changes because the time-varying component of the geoid–ellipsoid separation is negligible over the time scales and precision of GNSS measurements (Wang and Soler 2014). Therefore, we directly use ellipsoidal height to quantify subsidence and uplift, bypassing the need for a geoid model.

Method for Estimating Site Velocities and Uncertainties from GNSS Data

In the GOM region, a greater number of GNSS stations have been continuously operating for more than a decade, providing critical data for analyzing secular crustal motion trends necessary for establishing a robust regional geodetic reference frame. However, accurately estimating site velocities (linear trends) from these long-term displacement time series is often challenging due to outliers, data gaps, abrupt steps, transitional steps, localized anomalies associated with groundwater pumping or drought events, and undocumented antenna, antenna mount, even antenna pole changes. Although GNSS data in the US portion of the GOM typically lack coseismic and postseismic effects, the aforementioned complications mean that simple linear fitting over the entire data set is insufficient for most GNSS time series and can potentially lead to erroneous velocity estimates.

To address these challenges, we implemented the artificial intelligence (AI)-driven site-velocity estimation method of Wang et al.

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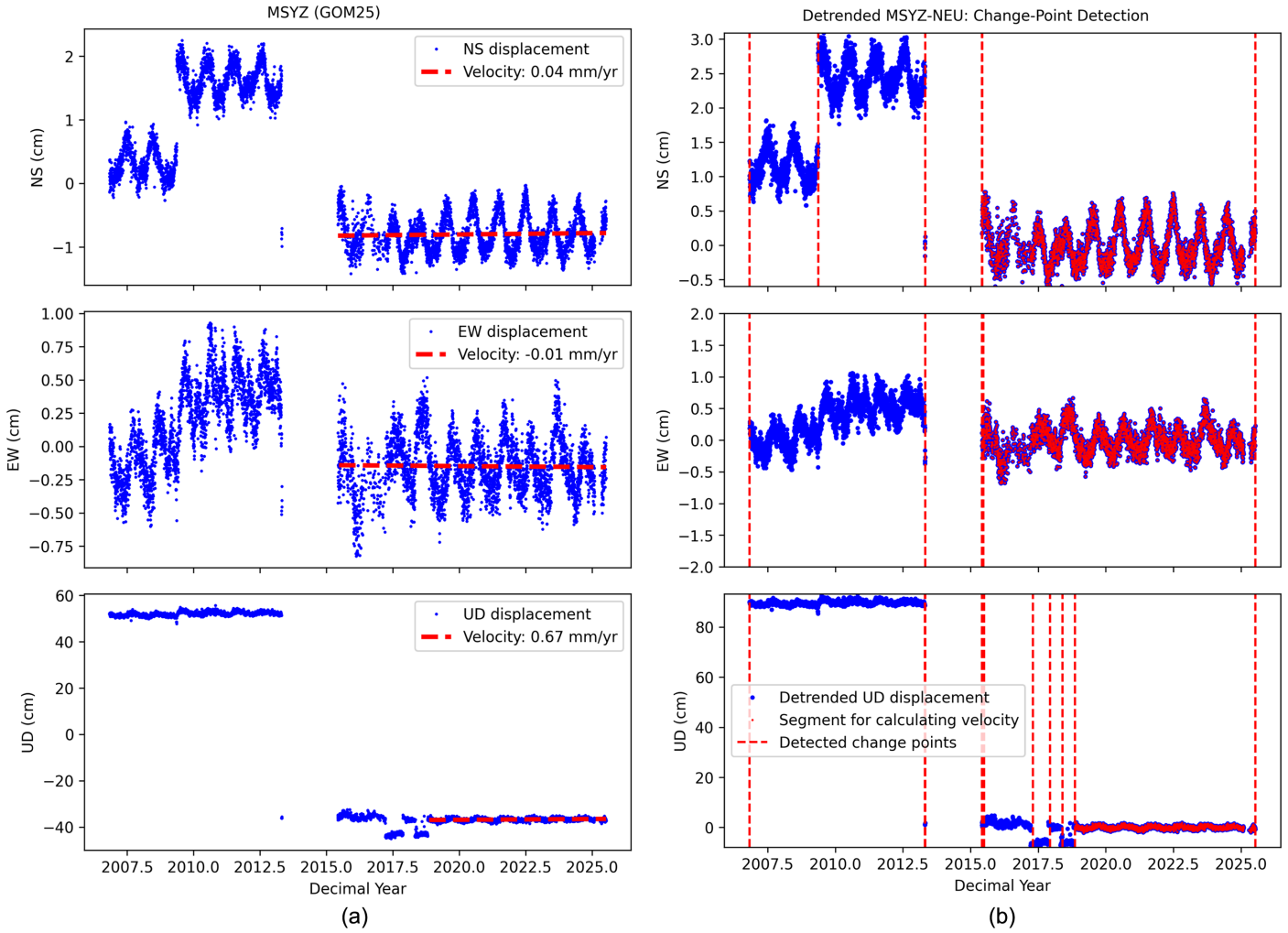


Fig. 2. Integrated analytical and AI methods for estimating site velocities: (a) estimation of site velocity from the longest change-point-free segment (minimum 4 years) for each component; and (b) detection of change points and isolation of change-point-free segments in detrended GNSS time series. Wang et al. (2025) has offered a detailed methodology.

(2025), which integrates a two-stage approach combining analytical and AI methods. The first stage uses analytical methods to identify potential change points—precise epochs marking anomalous steps in the data, including both instantaneous displacements (jumps) and transitional changes (slow drifts)—by evaluating various step thresholds. However, when processing time series from hundreds to thousands of stations, applying a unified threshold is infeasible because optimal values vary across regions and even between the two horizontal components at a single station.

Therefore, the second stage employs a convolutional neural network (CNN) model, trained on over 6,000 GNSS time series plots, to visually assess the suitability of these candidate thresholds for site velocity estimation. This stage iteratively refines the analytical parameters and determines the optimal configuration of thresholds for both instantaneous and transitional steps, ensuring reliable velocity estimates. Finally, the site velocity for each component is calculated by applying linear regression to the longest change-point-free segment of the GNSS time series, with a minimum duration of 4 years.

Fig. 2 illustrates the AI-driven workflow used to select change-point-free segments and estimate site velocities, demonstrated at a typical long-history GNSS station in the GOM region. MSYZ (Yazoo City, Mississippi) is a continuous GNSS station installed on the roof of a 1-story office building. It has ∼18 years of operation, featuring a long data gap and multiple equipment/mount changes. Documented antenna/receiver changes from the site log were already modeled in the PPP processing.

Fig. 2(a) shows the displacement time series in the GOM25 frame and the best-fit linear trends over the selected change-point-free segments. Fig. 2(b) shows the detrended three-component displacement time series with change points detected by the change-point CNN and the change-point-free segment for velocity estimation. Despite several discontinuities and a data gap longer than 1 year, the method isolates a ∼10-year change-point-free segment for the NS and EW components and a ∼7-year segment for the UD component. These examples demonstrate how CNN screening identifies the longest change-point-free segment, enabling defensible site velocity estimates even in records with large gaps and frequent undocumented changes. Using different segment lengths for different components is therefore statistically valid and maximizes the use of change-point-free data for each direction, yielding velocity estimates with higher confidence than forcing a common—and often shorter—segment across all components.

The requirement for a minimum 4-year change-point-free data segment ensures the statistical reliability of our site velocity estimates. As demonstrated by an analysis of ∼9,700 global GNSS stations,

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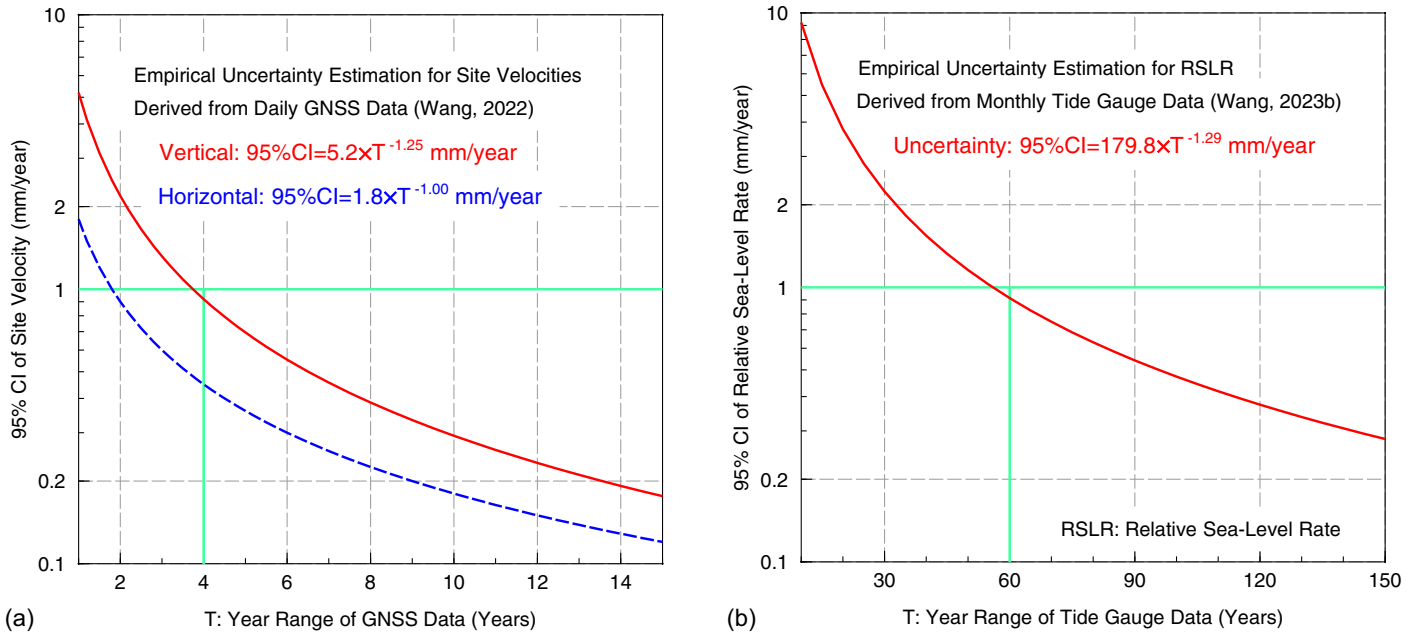


Fig. 3. Empirical models showing how the 95% CI of a linear trend derived from a time series depends on observation length: (a) 95% CI for vertical and horizontal site velocities derived from daily GNSS positional (or displacement) time series [after Wang (2022)]; and (b) 95% CI for relative sea-level change rates derived from monthly tide gauge data [after Wang (2023b)].

the 95% confidence interval (CI) of velocity estimates decreased exponentially with observation length [Fig. 3(a)] (Wang 2022). Generally, the horizontal component exhibited lower uncertainty in site velocity estimates compared with the vertical component, given the same duration of GNSS data. The 95% CI for site velocity estimates dropped below 1 mm/year after approximately 4 years for all three components. Therefore, a 4-year minimum guarantees submillimeter per year uncertainties for resolving regional tectonic motions and anthropogenic deformation like land subsidence. Although longer observations further reduced uncertainty, the improvement beyond about 7 years was marginal [Fig. 3(a)].

Realization of GOM25

General Methodology

To establish a stable regional geodetic reference frame that reveals local deformation signals, a fundamental requirement is that a selected set of reference stations exhibits, on average, NNT and NNR relative to the stable interior of the tectonic block. This condition ensures that local velocity trends—such as those from faulting, subsidence, or landslides—are not obscured by the broader-scale rigid-body motion inherent in the global reference solution. In practice, the regional frame is anchored to a well-established global frame through a set of reference stations that have precisely determined coordinates and velocities in both systems. A classic seven-parameter Helmert transformation (three translations, three rotations, and one scale factor) is applied, enforcing NNT and NNR constraints to ensure that the reference stations exhibit minimal translational and rotational residuals in the resulting regional frame.

Defining the Reference Network: Inclusion and Exclusion Criteria

An essential step in defining a regional geodetic reference frame is establishing rigorous inclusion and exclusion criteria for reference stations. For the previous GOM20 frame, we utilized a set of 55 long-term GNSS stations strategically distributed across the US portion of the GOM Coastal Plain. For GOM25, we expanded this network to the stable interior of the North American Plate within the northern GOM region, including 150 continuous GNSS stations, as depicted in Fig. 1. GNSS stations in the southern portion of the GOM (Mexico, Cuba, and Central America) are often characterized by short operational histories, frequent data gaps, and localized instability associated with postseismic displacement or faulting activity. These stations were therefore excluded from the reference network to ensure the long-term stability and continuity required for defining a robust regional frame.

The GOM25 reference network extends from the Atlantic Coast in the east to the western margin of the High Plains and Permian Basin in the west (Fig. 1). The High Plains region is well known to be experiencing steady minor uplift relative to the stable interior of the North American Plate (e.g., Overacker et al. 2022; Bogolub and Jones 2022). Incorporating stations from this uplifting region would introduce a nonrepresentative vertical motion, directly conflicting with the objective of anchoring the frame to the stable cratonic behavior of the northern GOM region. Similarly, stations within the Permian Basin were deliberately excluded due to potential instability that could be caused by subsurface fluid extraction (oil, gas, and groundwater) (e.g., Staniewicz et al. 2020; Kim and Lu 2021) and injection (e.g., Graves et al. 2023).

To ensure a stable datum, we restricted the GOM25 reference set to stations outside the Gulf Coast Aquifer—a major aquifer system that parallels the GOM coastline from Louisiana to the Mexican border (Casarez 2020) (Fig. 1). The system includes the Jasper, Evangeline, and Chicot aquifers, composed of discontinuous sand, silt, clay, and gravel. Because this reservoir is heavily used for municipal, agricultural, and industrial supply, intensive withdrawal and episodic recharge can drive substantial vertical land motion (e.g., Ellis et al. 2023; Turco et al. 2025), making sites within the aquifer unsuitable as long-term stable references. Constraining the reference network to locations beyond the aquifer area strengthens

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the conceptual integrity of the frame and improves its long-term stability for resolving groundwater-related deformation.

We also scrutinized stations in the Mississippi Alluvial Plain (MAP) (Fig. 1), an agricultural lowland extending from Illinois to Louisiana, covering approximately 116,500 km^2 (Painter and Westerman 2018). The MAP is underlain by the Mississippi River Valley alluvial aquifer, a shallow system that sustains intensive irrigation and has undergone long-term groundwater-level declines since the late 1920s. Increased withdrawals threaten further depletion, leading to widespread subsidence and occasional elastic uplift tied to recharge and sediment unloading (e.g., Wolstencroft et al. 2014; Reba et al. 2017; Van Arsdale et al. 2019). Consequently, many GNSS stations within or adjacent to the MAP exhibit vertical instability. For realizing GOM25, we therefore avoided using MAP stations as references to minimize bias in the regional frame, even though a subset of these stations exhibited apparent stability.

In summary, we initially selected stations anchored to the stable interior of the North American Plate (excluding areas with known deformation) with a minimum of 4 years of change-point-free data. This preliminary screening yielded approximately 500 candidate stations. We then applied an iterative screening process during the determination of the Helmert parameters, progressively tightening the residual threshold from a coarse cutoff (e.g., >1 cm/year) to a final value of 1 mm/year in any component [east-west (EW), north-south (NS), or upper-down (UD)]. Stations exceeding the current threshold at each iteration—indicative of instability or local deformation—were removed to preserve frame integrity. After accounting for data quality and spatial distribution, 150 stations were selected as the final reference network for GOM25 (Fig. 1). The vast majority of the reference stations now provide over 7 years of change-point-free continuous data.

The expanded network reflects improved data availability and spatial coverage compared with the previous GOM20, substantially strengthening the stability and utility of the regional reference frame. Although the 150-station network may seem overly dense, it ensures critical resilience. Subsampling tests confirmed that the estimated Helmert parameters are robust. Thinning the network to 100 stations (a one-third reduction) yielded only marginal changes to the transformation parameters. These changes, in turn, introduced negligible differences into the final site velocities across the GOM region.

The current redundancy therefore offers two clear benefits: greater numerical stability and resilience to undetected local site motion through the inclusion of a dense network of high-quality, long-record stations; and direct support for local-scale GNSS analyses using double-difference (DD) positioning, a method critical for land surveying and structural health monitoring. Because identifying a stable local reference is paramount for such applications, practitioners can confidently select the nearest rigorously screened GOM25 reference stations. Importantly, this redundancy does not impose additional burdens on end users of the regional reference frame. Users only need the seven parameters for XYZ coordinate transformation between IGS20 and GOM25, without any need to manage or select individual reference stations.

Frame Stability

Understanding the stability of a geodetic reference frame is essential prior to its practical application, yet it is a nontrivial task due to the lack of a single, universally accepted metric. For GOM25, we quantify stability by the root-mean square (RMS) of the residual velocities of its 150 reference stations, which are 0.3 mm/year (EW), 0.3 mm/year (NS), and 0.5 mm/year (UD). These RMS values, together with average residual velocities below 0.05 mm/year in all components, indicate that the reference network behaves as a stable rigid block for regional engineering applications (e.g., infrastructure monitoring and coastal hazard assessment).

The accuracy (uncertainty) of site velocities in the GOM25 frame is governed by both the trend uncertainty from GNSS time series analysis and the stability of the reference frame itself. For change-point-free segments with a minimum duration of 4 years (and an average of approximately 13 years), the 95% CIs for GNSS-derived linear trends are below 0.5 mm/year in both horizontal and vertical directions [Fig. 3(a)]. The stability of the GOM25 frame contributes an additional uncertainty below 0.5 mm/year. Accordingly, as a conservative, regional-scale move/no-move threshold for maps encompassing hundreds of stations, we adopted ± 1 mm/year in all three components. This value represents the practical detectability limit of the current GNSS geodetic infrastructure within the GOM region. However, velocities below this threshold are not necessarily devoid of physical meaning. Consistent, minor positive or negative velocities observed over a local or regional scale may reflect genuine geophysical phenomena.

Transformation Methodology: Between IGS20 and GOM25

One-Set Seven-Parameter Formulation

In geodesy, two broad strategies are used to transform GNSS position time series between reference frames: (1) a daily seven-parameter approach, and (2) a time-independent 14-parameter approach, both based on the Helmert transformation between Cartesian coordinate systems.

The daily seven-parameter method calculates a set of Helmert transformation parameters (three translations, three rotations, and one scale) for each day to align that day's coordinates in one frame to another. It has been adopted by, for instance, UNAVCO to produce daily solutions for the Plate Boundary Observatory (PBO) (Herring et al. 2016), as well as by the NGL for their daily solutions referenced to the North American Reference Frame (NA12) (Blewitt et al. 2013). Although effective, computing daily transformations can be cumbersome for many end-users who require straightforward, long-term stable reference frames. To address this, the geodetic community developed a time-independent 14-parameter transformation model, comprising the seven initial Helmert parameters and their constant rates of change. Prominent implementations include the NAD 83 realizations (2011, PA11, and MA11), tied to ITRF2008 via 14-parameter transformations (Pearson and Snay 2013).

The 14-parameter method is based on the assumption of the linearity of the tectonic plate motion, and thus the linear changes of these seven parameters over time. Over the last decade, the 14-parameter model has been further simplified to a one-set seven-parameter formulation that aligns two reference frames exactly at a chosen epoch t_0 (Wang et al. 2013). Specifically, we impose two conditions. First, at the alignment epoch t_0 (e.g., 2,025.0), every reference station has identical ECEF-XYZ coordinates in both the global and regional frames; consequently, the instantaneous Helmert parameters at t_0 are zero.

Second, because the reference stations are defined to have zero velocity in the regional frame, their coordinates in the regional frame remain constant—equal to their values at t_0—for all times. Thus, at a second epoch t_1 (e.g., 2,020.0), each station's regional-frame coordinates are still the same as at t_0, and its global-frame coordinates are known from the PPP solutions. In practice, the XYZ coordinates at t_0 and t_1 are represented by the 30-day mean around each epoch. Thus, the conventional seven Helmert transformation at epoch t_1 can

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Table 1. One-set seven parameters for realizing GOM25

Parameters[a]	Units	IGS20 to GOM25
t_0	year	2025.0
T'_x	m/year	1.66091724946714E-003
T'_y	m/year	−1.97334094972856E-004
T'_z	m/year	1.00010460090525E-003
R'_x	rad/year	−5.77734202536379E-11
R'_y	rad/year	−3.05098219529021E-09
R'_z	rad/year	−2.09858539285090E-10

[a]These seven parameters are used to transform ECEF-XYZ coordinates from IGS20 to GOM25 according to Eq. (3). A positive rotation value denotes a counterclockwise rotation of GOM25 relative to IGS20, viewed from the positive axis toward the origin.

be calculated. Because the seven parameters at epoch t_0 are zero, the rates of these seven parameters can be derived. The transformation is then described entirely by these rates: translation rates T'_x, T'_y, and T'_z; rotation rates R'_x, R'_y, and R'_z; and a scale rate s'. Because both frames are realized from the same GNSS measurements, no interframe scale change is expected; therefore $s' = 0$. The model thus reduces to six rate parameters plus the alignment epoch t_0. We refer to this as the one-set seven-parameter method (t_0, T'_x, T'_y, T'_z, R'_x, R'_y, and R'_z) for transforming positional time series.

This approach preserves the time dependence of frame divergence without day-to-day recalculations and retains seasonal and nonlinear signals present in the original time series, whereas daily seven-parameter realignments can partially remove such common signals at regional scales. This one-set seven-parameter transformation greatly simplifies use: a fixed parameter set lets users transform multiyear position (or displacement) time series with one formula, eliminating daily re-estimation. It is well-suited for long-term studies of subsidence, faulting, and other deformation processes.

This approach in has previously been implemented and documented in the realization of several stable regional reference frames (e.g., Kearns et al. 2019; Agudelo et al. 2020; Bao et al. 2021). The detailed method for calculating the one-set seven parameters has been given by Wang et al. (2020). These seven parameters for transforming ECEF-XYZ time series from IGS20 to GOM25 are listed in Table 1. In the transformation from IGS20 to GOM25, a positive rotation value indicates a counterclockwise rotation of the regional frame relative to the global frame when viewed from the positive end of the corresponding axis toward the origin.

Justification of the Velocity-Only Approach for Reference Stations

Modern trajectory models have evolved to encompass secular trends, seasonal oscillations, and transient deformations (e.g., Bevis and Brown 2014; Sobrero et al. 2020). These advancements are now integrated into global reference frames as part of an augmented kinematic model that accounts for nonlinear station motions, including annual and semiannual seasonal oscillations and postseismic deformation (Altamimi et al. 2023). For the definition of a stable regional reference frame like GOM25, however, the objective is to capture the long-term linear motion of the tectonic plate interior to serve as a stable datum, not model every temporally varying signal at each reference station. Consequently, we adopt a simplified velocity-only approach for the reference stations. This is justified on three grounds.

First, the northern GOM region is characterized by seismic quiescence; our selected reference stations, specifically chosen from the stable interior of the North American Plate, exhibit no coseismic or postseismic signals in the two-decade observation period. Therefore, complex trajectory models are unnecessary for defining the long-term behavior of our reference network.

Second, the primary application of GOM25 is to serve as a stable foundation for delineating nonlinear short-term signals—such as seasonal hydrological or anthropogenic deformation—in the broader regional network. Overmodeling the reference stations by removing these seasonal signals would effectively remove the variability that end-users seek to measure relative to a stable datum. Our approach preserves these signals in the final products for users.

Third, methods that emphasize seasonal modeling of site deformation in realizing reference frames are primarily designed to ensure that regional frames inherit the full kinematic behavior—including periodic signals—of the global frame, thereby maximizing formal consistency (Gómez et al. 2022). Although valuable for maintaining spectral alignment between frames, this priority differs from the objective of GOM25. Our goal is to create a frame that optimally isolates local deformation by removing only the broad, linear plate motion, leaving seasonal and other nonlinear signals intact for subsequent geophysical and engineering analyses. A velocity-only description of the reference stations is sufficient to achieve this goal in a tectonically quiet region.

Therefore, applying a simple linear regression to the longest continuous change-point-free segment of each reference station's time series (with a minimum duration of 4 years) is not only sufficient but optimal for realizing a frame that is both kinematically stable and maximally useful for the study of local and temporal signals in the GOM region.

Transforming ECEF-XYZ Coordinates

GOM25 is aligned in origin and scale with IGS20 at a single alignment epoch t_0; no scale-rate is used. Let

$$P(t) = \begin{bmatrix} X(t) \\ Y(t) \\ Z(t) \end{bmatrix}; \qquad T' = \begin{bmatrix} T'_x \\ T'_y \\ T'_z \end{bmatrix}; \qquad R' = \begin{bmatrix} R'_x \\ R'_y \\ R'_z \end{bmatrix}; \qquad \Delta t = t - t_0$$

where T'_x, T'_y, and T'_z = translation rates (m/year); and R'_x, R'_y, and R'_z = rotation rates (rad/year). With the one-set seven-parameter method, the forward transformation the ECEF-XYZ position from IGS20 to GOM25 is

$$P_{\mathrm{GOM25}}(t) = P_{\mathrm{IGS20}}(t) + \Delta t T' + \Delta t [R']_{\times} P_{\mathrm{IGS20}}(t) \tag{1}$$

where the skew-symmetric cross-product matrix $[R']_{\times}$ is

$$[R']_{\times} = \begin{bmatrix} 0 & R'_z & -R'_y \\ -R'_z & 0 & R'_x \\ R'_y & -R'_x & 0 \end{bmatrix} \tag{2}$$

Explicitly

$$\begin{aligned} X(t)_{\mathrm{GOM25}} &= X(t)_{\mathrm{IGS20}} + T'_x \cdot (t - t_0) + R'_z \cdot (t - t_0) \cdot Y(t)_{\mathrm{IGS20}} \\ &\quad - R'_y \cdot (t - t_0) \cdot Z(t)_{\mathrm{IGS20}} \\ Y(t)_{\mathrm{GOM25}} &= Y(t)_{\mathrm{IGS20}} + T'_y \cdot (t - t_0) - R'_z \cdot (t - t_0) \cdot X(t)_{\mathrm{IGS20}} \\ &\quad + R'_x \cdot (t - t_0) \cdot Z(t)_{\mathrm{IGS20}} \\ Z(t)_{\mathrm{GOM25}} &= Z(t)_{\mathrm{IGS20}} + T'_z \cdot (t - t_0) + R'_y \cdot (t - t_0) \cdot X(t)_{\mathrm{IGS20}} \\ &\quad - R'_x \cdot (t - t_0) \cdot Y(t)_{\mathrm{IGS20}} \end{aligned} \tag{3}$$

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where $X(t)_{IGS20}$, $Y(t)_{IGS20}$, and $Z(t)_{IGS20}$ = XYZ coordinates (at epoch t) of a site with respect to IGS20 (m); $X(t)_{GOM25}$, $Y(t)_{GOM25}$, and $Z(t)_{GOM25}$ = XYZ coordinates of the site with respect to GOM25 (m); t_0 = epoch that aligns the coordinates with respect to these two reference frames, which is fixed at epoch 2025.0 (year) for GOM25, where a site has identical XYZ coordinates with respect to IGS20 and GOM25 at t_0; and T'_x, T'_y, T'_z, R'_x, R'_y, and R'_z = constant parameters indicating the rates (one-time derivates) of translational shifts and rotations along the x-, y-, and z-coordinate axes from IGS20 to GOM25 (Table 1).

The inverse transformation from GOM25 to IGS20 is

$$P_{IGS20}(t) = (I + \Delta t[R']_\times)^{-1}(P_{GOM25}(t) - \Delta t T') \tag{4}$$

where I is the 3×3 identity matrix. For typical geodetic applications characterized by small rates and multidecade time spans, $(I + \Delta t[R']_\times)^{-1}$ can be approximated as $(I - \Delta t[R']_\times)$. This yields the sufficient first-order form as follows:

$$\begin{aligned} X(t)_{IGS20} \approx\ & X(t)_{GOM25} - T'_x \cdot (t - t_0) - R'_z \cdot (t - t_0) \\ & \cdot Y(t)_{GOM25} + R'_y \cdot (t - t_0) \cdot Z(t)_{GOM25} \\ Y(t)_{IGS20} \approx\ & Y(t)_{GOM25} - T'_y \cdot (t - t_0) + R'_z \cdot (t - t_0) \\ & \cdot X(t)_{GOM25} - R'_x \cdot (t - t_0) \cdot Z(t)_{GOM25} \\ Z(t)_{IGS20} \approx\ & Z(t)_{GOM25} - T'_z \cdot (t - t_0) - R'_y \cdot (t - t_0) \\ & \cdot X(t)_{GOM25} + R'_x \cdot (t - t_0) \cdot Y(t)_{GOM25} \end{aligned} \tag{5}$$

In practice, users typically work with local ENU displacement time series derived from IGS20 solutions. These can be transformed into the GOM25 frame by applying the one-set seven-parameter transformation to their ECEF-XYZ coordinates as an intermediate step. The detailed methodology for transforming ENU displacement time series and site velocities between IGS20 and GOM25, along with a suite of open-source Python tools that implement these transformations, have been provided by Wang and Welch (2026) to facilitate practical adoption of GOM25.

Assessing the Spatial Applicability of GOM25

Euler Pole Estimation

In practice, the transformation from a global frame to a regional frame is dominated by a small rigid rotation, with translation rates typically on the order of a few millimeters to submillimeters per year (Table 1). The motion of a rigid tectonic block within a global reference frame can be described as a rotation about an axis passing through Earth's center of mass; the point where this axis intersects the surface is the Euler pole. Conceptually, the transformation from IGS20 to GOM25 removes the large-scale rigid-body motion of the North American Plate by first removing the average translational motion of the reference stations and then applying a counter-rotation (the negative of the Euler vector) to the residual velocities. In practice, the Helmert parameters provided in Table 1 implement both the translation and rotation components simultaneously.

The Euler pole location and the rotation rate can be estimated according to the three rotation rates R'_x, R'_y, and R'_z used to realize the regional reference frame (Table 1)

$$\lambda_p = \text{atan2}(R'_y, R'_x) \tag{6}$$

$$\varphi_p = \tan^{-1}\left(\frac{R'_z}{\sqrt{R'^2_x + R'^2_y}}\right) \tag{7}$$

$$\omega = \sqrt{R'^2_x + R'^2_y + R'^2_z} \tag{8}$$

where λ_p (longitude) and φ_p (latitude) = location of the Euler pole (rad); and ω = rotation rate (rad/year). A positive longitude estimate (λ_p) indicates east of the prime meridian, and a negative longitude estimate indicates west of the prime meridian. A positive latitude estimate (φ_p) indicates north of the equator, and a negative latitude estimate indicates south of the equator. By convention, a positive ω corresponds to a counterclockwise rotation when viewed from the pole toward the origin (Earth's center). To visualize this on a map for a horizontal rotation, a positive rotation (e.g., of the North American Plate) would be counterclockwise, meaning a site would move toward the west/southwest relative to a fixed Euler pole.

These three rotation rates, in conjunction with the three translation rates, serve to mathematically minimize the residual motion of the selected reference network in the regional frame; they do not necessarily define the unique physical Euler pole of a tectonic block. Based on our analysis, small differences in the selection of reference stations can yield noticeable changes in the inferred pole position; moreover, distinct Euler poles can produce nearly indistinguishable velocity fields within the footprint of the reference network.

The horizontal speed induced by rotation about the Euler pole at a site separated by great-circle angle θ from the pole is given by

$$v = \omega D \sin(\theta) \tag{9}$$

where D = Earth's mean radius (m). For a positive (counterclockwise) rotation ω, the induced horizontal velocity v is tangential to the small circle around the Euler pole and points in the direction of local plate motion, which is westward for the North American Plate in a global reference frame. For a tectonic plate of finite size, the angular distance (θ) of any point on the plate from the Euler pole is necessarily less than $\pi/2$. Therefore, the value of $\sin(\theta)$ increases with distance from the pole across the entire plate, and the rotational velocity (v) always increases with distance from the Euler pole.

For GOM25, the derived rotation rates imply an Euler pole at 91.085° W; 3.934° S, roughly 40° (θ) south of the core area of the GOM25 reference network. The transformation from IGS20 to GOM25 can be realized by a clockwise rotation of 0.175° per million years (Myr) about this pole, combined with minor translational adjustments. This clockwise rotation represents the negative of the counterclockwise motion of the southern North American Plate observed in IGS20. The pole is located in the eastern Pacific Ocean, near the Galapagos Islands and is approximately 4,000 km from the center of the reference network (Fig. 4). Its position and rotation rate are similar (within a few degrees of longitude and latitude) to published Euler vectors for the North American Plate within global reference frames IGS08 and IGS14 (e.g., Blewitt et al. 2013; Kreemer et al. 2018; Ding et al. 2019), as well as the Euler vector for the North American Terrestrial Reference Frame of 2022 (NATRF2022) within ITRF20 (≈86.298° W; 3.821° S, and 0.197°/Myr) (Goudarzi et al. 2026).

In Fig. 4, vectors are not shown for stations with horizontal velocities below 4 mm/year, but their velocities are distinguished by distinct marker symbols. The location of the GOM25 Euler pole relative to IGS20 is also shown. Plate boundaries are from the NNR-MORVEL model (DeMets et al. 2010), with a modification to the southeastern North American–Caribbean boundary. Based on the GOM25 velocity field, we reposition this boundary from

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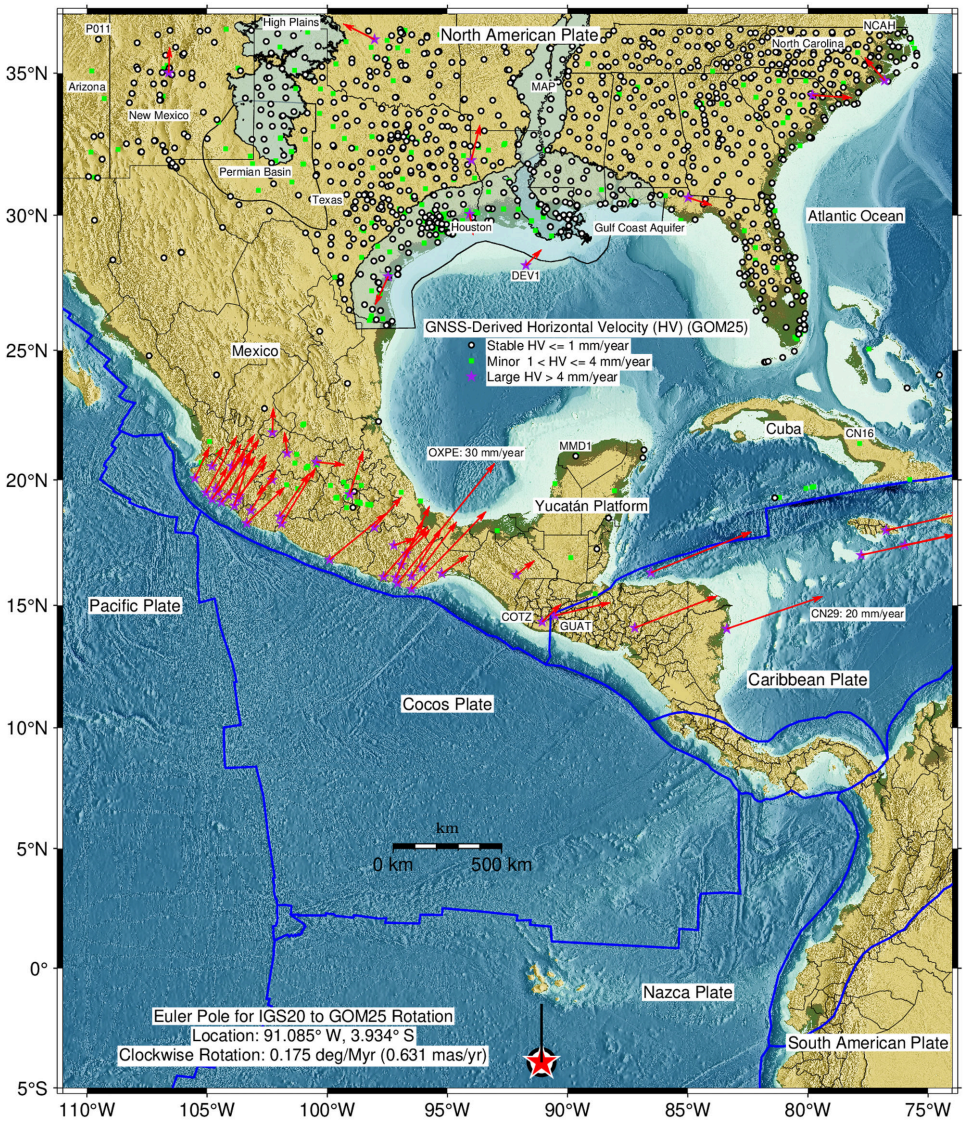


Fig. 4. Horizontal velocity vectors at GNSS stations with respect to GOM25. Plate boundaries are from the NNR-MORVEL model (DeMets et al. 2010), with a modification to the southeastern North American–Caribbean boundary. [Topo-bathymetric data from GEOBCO Compilation Group 2024; base map generated using The Generic Mapping Tools, v 6.3.0 (Wessel et al. 2019).]

its original location south of station GUAT to a position between stations COTZ and GUAT, which are approximately 65 km apart.

Spatial Domain of Applicability

Fig. 5 compares the vertical velocities of the reference stations in the IGS20 and GOM25 frames. In IGS20, the stations showed a regional subsidence signal (average −0.7 mm/year), representing common-mode motion within the global frame rather than local deformation. Because nearly all stations exhibit downward velocities, cancellation is minimal and the average reliably quantifies this net vertical motion of the North American Plate relative to the geocenter. GOM25 successfully removes this regional bias, reducing the average vertical velocity to near zero (0.01 mm/year) with a standard deviation of 0.48 mm/year. Within the core network area, all reference stations exhibit residual velocities below 1 mm/year, confirming that the Helmert transformation (translational adjustments and counterrotation) effectively removed the large-scale rigid-body motion of the southern North American Plate.

Farther outside this defined domain (Fig. 5), extrapolating the Euler-pole rotation could introduce spurious velocities (too large or too small) that do not correctly reflect local deformation. To evaluate the limits of the GOM25 transformation, we examined the displacement time series at four GNSS stations (P011, NCAH, MMD1, and CN16) located far beyond the reference network footprint. Locations of these stations are shown in Fig. 4. P011 (Arizona) and NCAH (North Carolina) are located in the stable interior of the North American Plate, approximately 5,000 km from the Euler pole. MMD1 (Yucatán Platform) and CN16 (Cuba) are situated on the stable southern margin of the plate, approximately 2,000 km from the Euler pole.

Fig. 6 demonstrates that all four sites exhibit steady velocities below 1 mm/year in the GOM25 frame, although station CN16 exhibits a slightly higher vertical velocity of 1.1 mm/year. The stability is consistent with their locations on stable crust. This observed stability is further supported by their adjacent stations (Fig. 4). Overall, approximately 90% of the ∼1,000 GNSS stations in the region show stability (horizontal velocities within ±1 mm/year) in GOM25 (Fig. 4). This confirms that the translational adjustments

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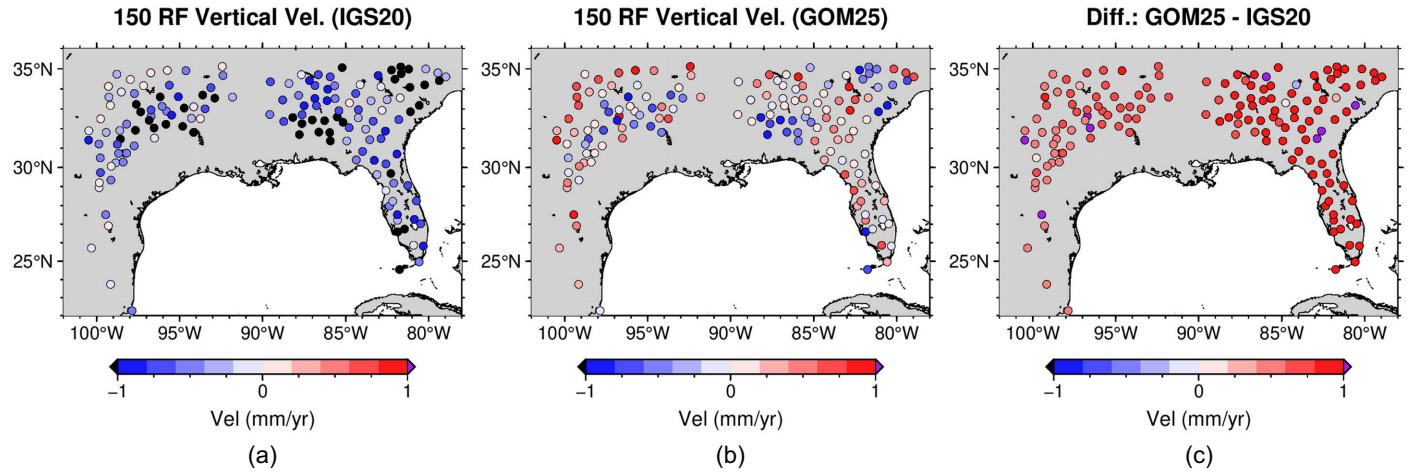


Fig. 5. Vertical ground velocities at the 150 reference GNSS stations used to realize the GOM25 reference frame: (a) velocity with respect to the global IGS20 frame; (b) velocity with respect to the regional GOM25 frame; and (c) velocity as the difference between the two frames (GOM25 minus IGS20). The stations show a mean downward velocity of −0.7 mm/year in IGS20 but a near-zero mean vertical velocity (0.01 mm/year) in GOM25.

and counterrotation accurately remove the large-scale rigid-body motion of the southern North American Plate. If the transformation were erroneous, these purportedly stable sites would display spurious velocities exceeding 1 mm/year.

In summary, the GOM25 reference frame is applicable for high-precision geodetic studies across the southern North American Plate, an extensive domain bounded by 15° to 37° N and 110° to 75° W. This area encompasses the stable core of the network—the US GOM coast—and extends effectively from central Mexico in the west to the Bahamas in the east, from New Mexico in the northwest to North Carolina in the northeast, and south to the Yucatán Platform and Cuba. Users can confidently employ GOM25 within this region to investigate fault slip, subsidence, uplift, plate boundary processes, and structural health monitoring. Extrapolation beyond these geographical limits is not recommended without independent verification of local stability. For studies extending beyond the recommended area, users should either (1) verify local stability by inspecting residual velocities against nearby stations, (2) re-estimate regional Helmert transformation parameters for a suitably expanded network, or (3) remain in the global frame (IGS20) for those areas.

Applications of GOM25

The GOM25 reference frame provides a unified geodetic foundation for the GOM region, enabling high-precision deformation monitoring across scales—from local faulting and urban subsidence to regional tectonic motion and sea-level rise—as well as the structural health monitoring of onshore and offshore infrastructure. Urban subsidence and faulting have been extensively investigated using its predecessor, the GOM20 frame, and this study focuses on regional-scale analyses enabled by the enhanced stability and spatial extent of GOM25.

Detecting Subtle Regional Uplift and Subsidence

Vertical land motion in the GOM region is driven by a mix of natural and anthropogenic processes. Natural contributors include sediment loading (e.g., Ivins et al. 2007), glacial isostatic adjustment (e.g., Peltier et al. 2015), and tectonic movement along growth faults paralleling the coast (e.g., Dokka et al. 2006). Anthropogenic activities, particularly fluid extraction, further exacerbate motion through aquifer compaction. The GOM25 reference frame effectively minimizes regional-scale vertical signals by removing an average of 0.7 mm/year of subsidence (Fig. 5).

Fig. 7 illustrates the vertical velocities, relative to GOM25, of approximately 1,000 GNSS stations within the recommended GOM25 application area. Generally, these vertical site velocities reflect the average rate of ground displacement from around 2010 to 2025. The majority of sites are stable (−1 to 1 mm/year), falling within the practical detection limit of current GNSS infrastructure. A consistent uplift signal of 0.5 to 2 mm/year was detected across the High Plains with respect to GOM25.

Fig. 8 depicts the three-component displacement time series for two long-history stations (TXAM: 1999–2025; P038: 2005–2025) anchored on the High Plains. Their locations are indicated in Fig. 7. Both stations exhibited sustained uplift of approximately 1.5 mm/year while maintaining horizontal stability (<0.3 mm/year). Our analysis of 45 long-history GNSS stations on the southern High Plains (Fig. 7) indicated an average uplift rate of 1.3 ± 0.6 mm/year with respect to GOM25. This estimate isolates the vertical signal by removing other regional common-mode vertical motions (e.g., GIA, plate-scale tectonic movements, and regional hydrological loading/unloading), yielding a value of 1.3 mm/year that differs from the ∼2 mm/year absolute uplift reported in studies using global reference frames (e.g., Overacker et al. 2022). The discrepancy highlights how our approach isolates the differential motion between the High Plains and the stable coastal plain.

Overacker et al. (2022) reported that the modern uplift at a couple of millimeters per year is spatially correlated with groundwater decline in the southern High Plains aquifer. This short-term anthropogenic signal is superimposed upon the long-term geological existence of the High Plains as an elevated province, which is attributed to deeper tectonic processes like dynamic mantle support (e.g., Liu and Gurnis 2010; Karlstrom et al. 2012) and isostatic adjustment to erosional unloading (e.g., McMillan et al. 2006). The high stability of the GOM25 reference frame now provides the necessary datum to resolve these modern motions clearly. Although a full discrimination between these short-term and long-term signals is beyond the scope of this methodological paper, GOM25 establishes a foundational capability for such targeted investigations in future work.

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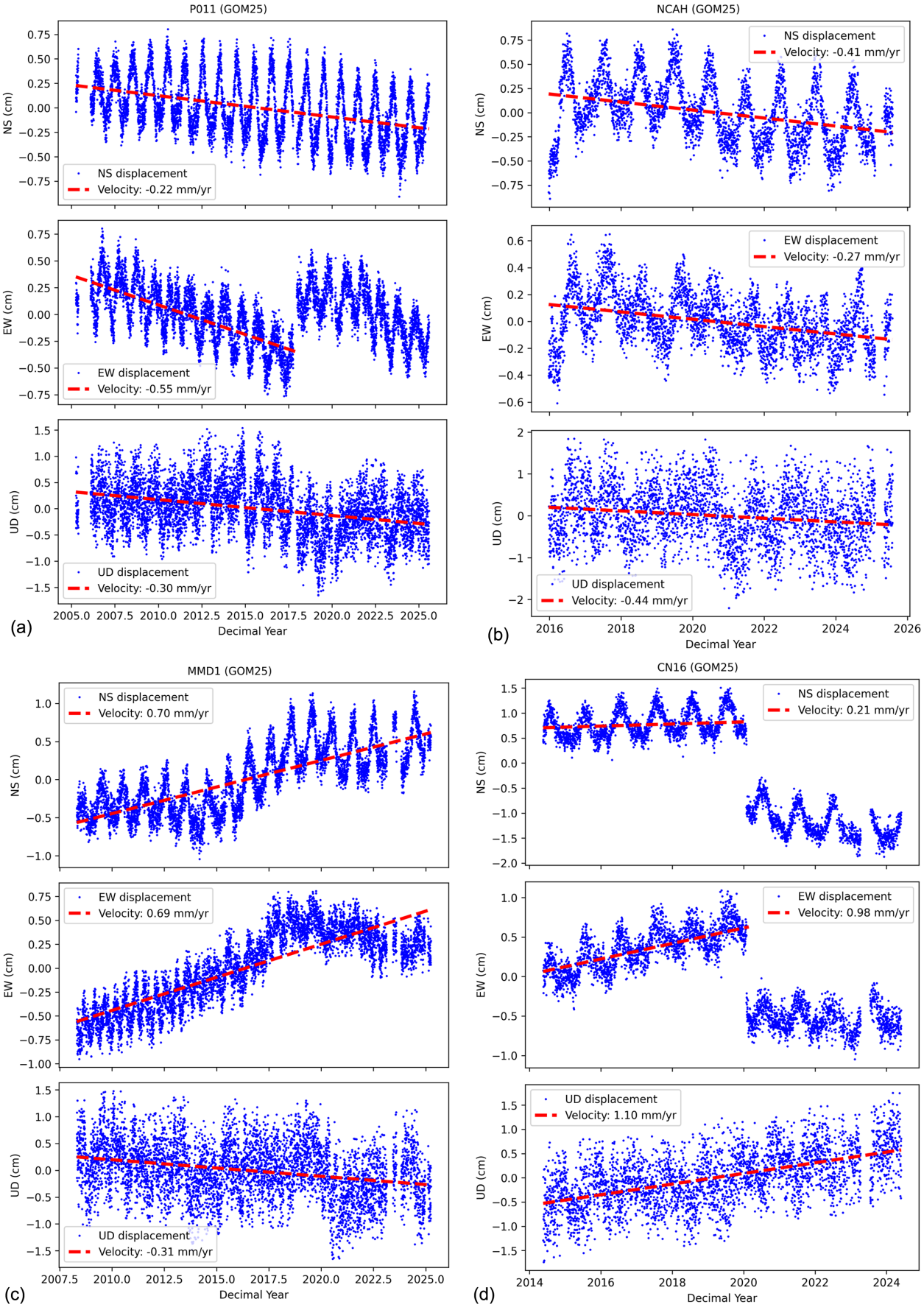


Fig. 6. Three-component displacement time series with respect to GOM25 for four GNSS stations (P011, NCAH, MMD1, and CN16) located far beyond the core reference network. All stations show velocities below 1 mm/year in all three components, except for CN16, whose vertical velocity is 1.10 mm/year.

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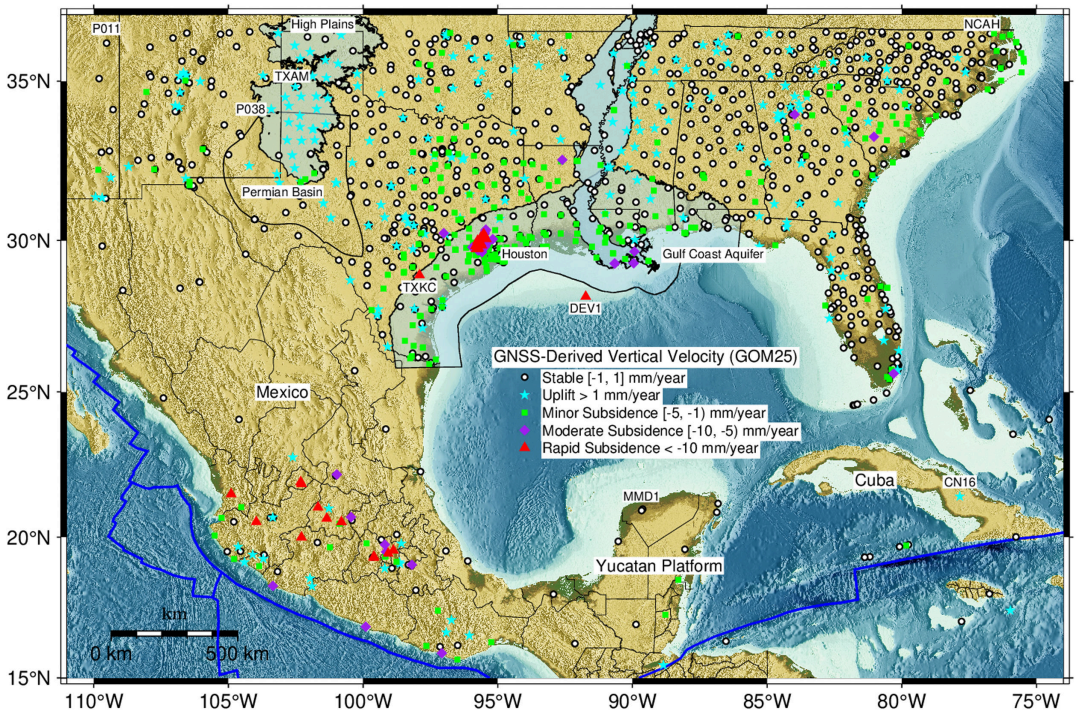


Fig. 7. Vertical land motion rates derived from GNSS data (approximately 2010–2025) within the GOM25 application domain (15°–37° N; 110°–75° W). Site velocities are differentiated [Topo-bathymetric data from GEOBCO Compilation Group 2024; base map generated using The Generic Mapping Tools, v 6.3.0 (Wessel et al. 2019).]

The most prominent subsidence patterns are observed in coastal zones and major aquifer systems. Widespread moderate subsidence (1–5 mm/year) is prevalent within the Gulf Coast Aquifer, the Texas Gulf Coastal Plain, the Mississippi Alluvial Plain, and along the Atlantic coast (Fig. 7). This distribution suggests a common driver, with groundwater extraction being the most likely primary cause, particularly because many subsiding sites coincide with urban and agricultural centers.

Along the coastal belt, natural consolidation of young sediments could also considerably contribute to ongoing subsidence. For example, in the Galveston, Texas, area, natural subsidence has been estimated at 2–4 mm/year (Zhou et al. 2021), and in other parts of the Texas coast, natural rates range from less than 2 mm/year. In the modern Mississippi Delta region, natural compaction of late Holocene to recent sediments reaches rates as high as 9 mm/year, as supported by GNSS and InSAR data and geologic studies (e.g., Törnqvist et al. 2008; Jankowski et al. 2017; Blum et al. 2023).

Although rapid subsidence (>10 mm/year) is largely confined to the well-documented regions of Houston and central Mexico (which were the focus of previous study using GOM20), two notable exceptions were identified outside the Houston metropolitan region: Sites TXKC and DEV1 (Fig. 9). Their locations are indicated in Fig. 7. TXKC is located near Karnes City, Texas. It exhibits a sustained subsidence rate of ∼17 mm/year beginning in 2014. There was no considerable subsidence at this site before 2014. This signal—characterized by pronounced vertical motion with minimal horizontal displacements (∼1 mm/year)—is indicative of aquifer-system compaction. This aligns with the hydrogeologic setting of compressible sediments and observed groundwater-level declines linked to intensive municipal, industrial, and agricultural groundwater withdrawals in the region (e.g., Haley et al. 2022; Qiao et al. 2023).

DEV1 is an offshore site installed on a fixed platform in 82 m of water, ∼130 km offshore of the Mississippi Delta. It records a persistent subsidence rate of ∼15 mm/year, about twice the typical subsidence rate for the delta front. This rapid subsidence likely reflects a combination of regional seabed subsidence and localized reservoir compaction associated with nearby hydrocarbon production (Wang 2023a). The DEV1 time series also underscores the value of GOM25 for the long-term structural health monitoring of offshore infrastructure, where traditional GNSS surveying methods using differential techniques are impractical due to the absence of nearby stable reference stations—baselines from onshore references are typically too long to maintain the required resolution.

In summary, the GOM25 vertical velocity field resolves three distinct deformation patterns across the region: (1) slight uplift (∼1.3 mm/year) in the southern High Plains; (2) widespread minor subsidence (1–5 mm/year) along the GOM and South Atlantic coasts, as well as in inland coastal plain areas, particularly in Texas, Louisiana, Mississippi, and South Carolina; and (3) rapid subsidence (>10 mm/year) in localized urban zones such as the greater Houston area and central Mexico. This detailed view enables refined hazard assessments and resource management for the Gulf Coast region.

Delineating Faulting and Interplate Motions

GOM25 provides a precise datum for isolating subtle intraplate deformation and clarifying motions across major plate boundaries. Within the stable interior of the North American Plate, horizontal velocity vectors (Fig. 4) revealed no coherent pattern of regional tectonic strain. Rates of 1–4 mm/year at isolated sites are attributable to localized processes—such as uneven subsidence or shallow fault creep (e.g., Engelkemeir and Khan 2007; Liu et al. 2019; Qu et al. 2019)—rather than to distributed plate-interior deformation. Thus, the GOM25 frame confirms that the Gulf Coast and its hinterland are tectonically quiet at the scale of a few millimeters per year.

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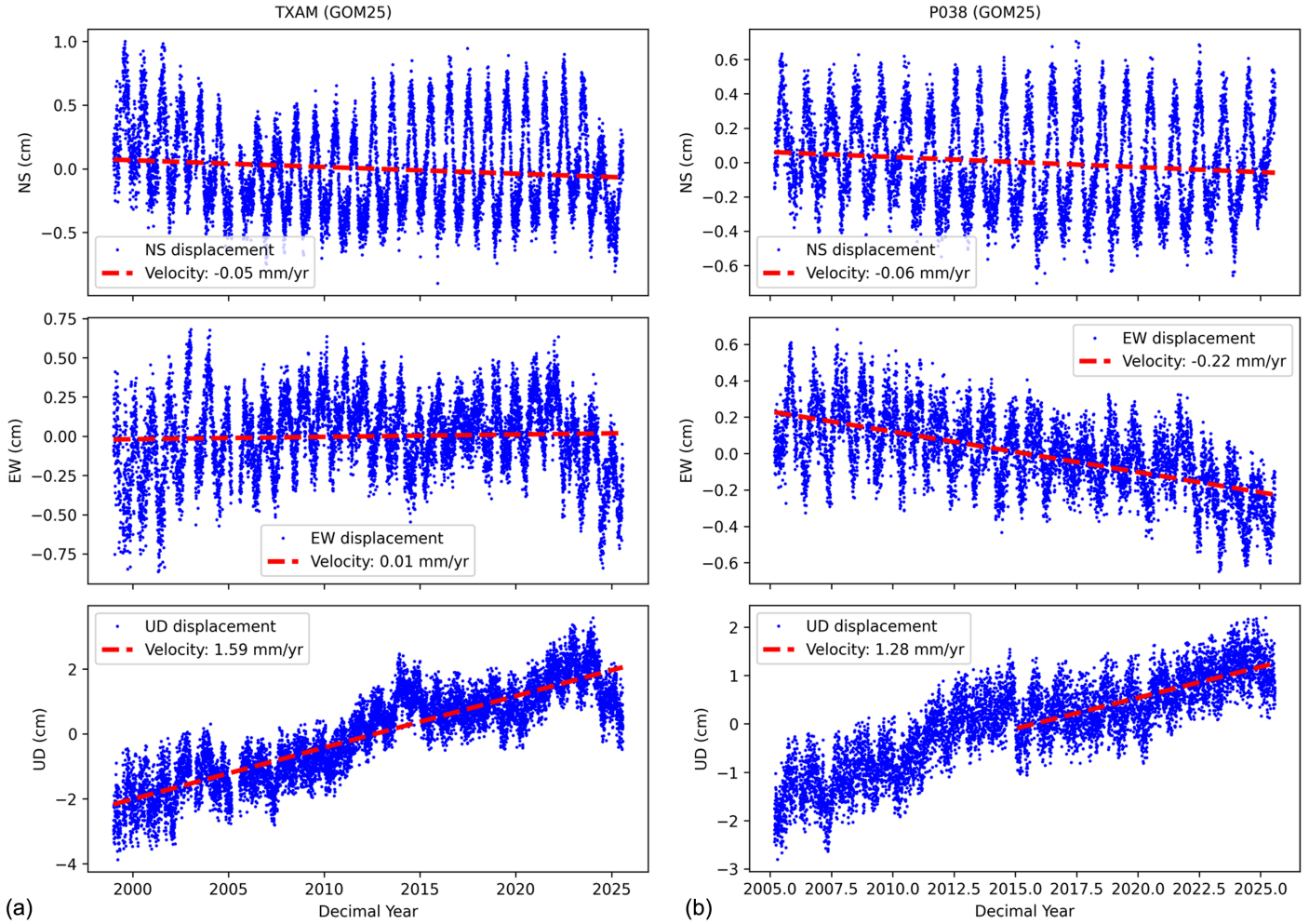


Fig. 8. Three-component displacement time series with respect to GOM25 for two long-term GNSS stations (TXAM, P038) on the High Plains. Both stations exhibit steady uplift of approximately 1.5 mm/year and horizontal velocities below 0.3 mm/year.

In contrast, GOM25 clearly resolves the high-strain signals along the plate margins. Along the southwestern boundary of the North American Plate, the subduction of the Cocos Plate drives velocities of up to 30 mm/year. To the southeast, a consistent ~20−mm/year left-lateral strike-slip motion is observed across the boundary between the North American (NA) and Caribbean (CA) plates.

The high stability of GOM25 enables a refined view of this complex plate boundary zone. Our results suggest a need to refine the precise location of the NA–CA plate boundary in Guatemala. Conventional plate models consistently placed both the COTZ (near Santa Lucía Cotzumalguapa) and GUAT (Guatemala City) GNSS stations on the NA plate (e.g., Bird 2003; DeMets et al. 2010; Argus et al. 2011). However, the horizontal velocity vectors with respect to GOM25 revealed a clear distinction: COTZ's velocity (both direction and magnitude) is consistent with nearby stations known to be on the stable NA plate. Conversely, GUAT's velocity is inconsistent with the NA plate but instead matches the motion of stations on the CA plate, suggesting it is part of that tectonic block.

GUAT and COTZ are located approximately 65 km apart within the Polochic-Motagua strike-slip fault system, which marks the on-land plate boundary between the NA and CA plates (e.g., Lyon-Caen et al. 2006; Garnier et al. 2022). This fault system is a complex crustal deformation zone, evolving plate boundaries where subduction, strike-slip motion, and magmatic activity interact near the NA–CA–Cocos triple junction (e.g., DeMets et al. 2000; Guzmán Speziale and Meneses-Rocha 2000; Parolari et al. 2025). Although it is not ideal to use a single line to represent a tectonic plate boundary, our results indicate that if a line must be drawn, it would be better placed between COTZ and GUAT, rather than south of GUAT (Fig. 4). This finding highlights how a high-stability regional reference frame can resolve details that are obscured in global solutions, ultimately leading to more accurate tectonic models.

Resolving Sea-Level Change Relative to the GOM Coast

A primary application of GOM25 is the precise determination of the sea-level rate (SLR) relative to the GOM coast, which requires isolating the change in sea surface height from the confounding effects of vertical land motion (VLM). The concept of SLR with respect to GOM25 is slightly different from the term absolute SLR, which is widely used in global-scale sea level studies. Absolute SLR typically refers to the change in the height of the sea surface relative to the Earth's center of mass—a fixed reference point known as the geocenter. For assessing long-term coastal flooding, inundation, and wetland change, the rate relative to GOM25 is the more relevant metric because it directly captures the changing relationship between sea level and the coastal landmass where infrastructure and communities are situated. A purely geocentric rate, although valuable for understanding ocean volume change

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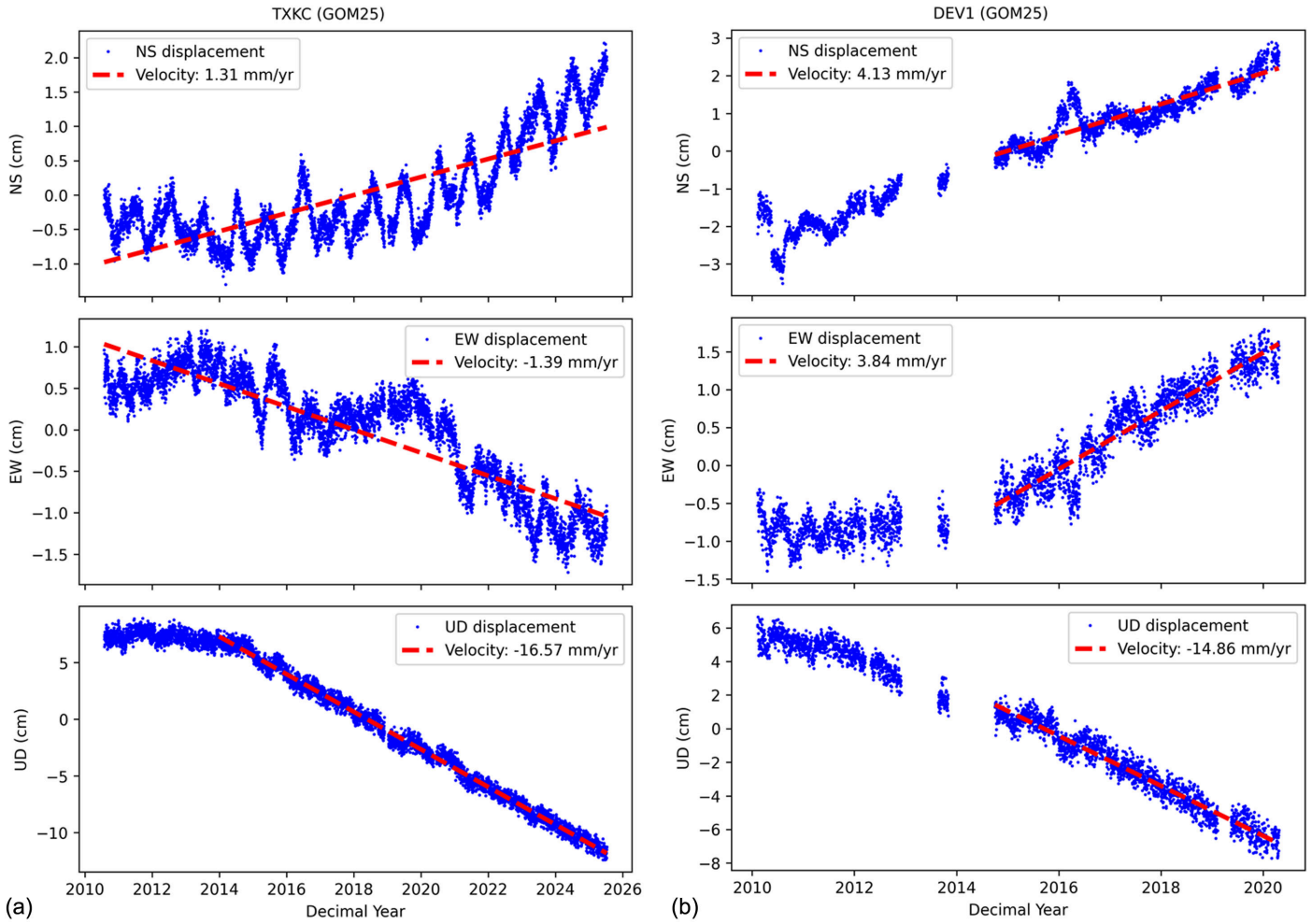


Fig. 9. Three-component displacement time series (2010–2025) with respect to GOM25 for two stations outside Houston that experienced rapid subsidence (>10 mm/year). TXKC is located in Karnes City, south-central Texas, and DEV1 is on an offshore platform in the Gulf of Mexico.

on a global scale, is less directly applicable to long-term coastal flooding assessment.

The original tide gauge sea-level measurements are referenced to local benchmarks, usually fixed to the same foundation as the tide-gauge sensor. The linear trend estimated from this monthly data is therefore a measure of relative sea-level rate (RSLR), which captures the combined effect of absolute sea-level change and local VLM. The SLR with respect to GOM25 can be calculated as follows:

$$\mathrm{SLR}_{\mathrm{GOM25}} = \mathrm{RSLR} + \mathrm{VLM}_{\mathrm{GOM25}} \tag{10}$$

In this formulation, a positive rate indicates land uplift or sea-level rise, and a negative rate indicates land subsidence or sea-level fall. This construction removes site-specific land motion and expresses sea-level change relative to GOM25, physically the stable coastal plain.

Because GOM25 is aligned to IGS20 at the reference epoch t_0 via a constant set of transformation parameters, velocities in GOM25 differ from those in IGS20 by a constant plate motion term. In the GOM region, a vertically stable site (zero velocity) with respect to GOM25 retains a vertical velocity of approximately -0.7 mm/year with respect to IGS20 (Fig. 5). The 0.7 mm/year reflects the motion of the North American Plate relative to the Earth's center of mass—primarily driven by a combination of GIA and regional plate tilt. For context, this magnitude is comparable to nearly one-third of the contemporary global mean sea-level rise rate ($\sim$2 to 3 mm/year), underscoring that plate-scale vertical motion is a nonnegligible component in sea-level budget calculations. Consequently, adding the VLM relative to GOM25 to the RSLR removes plate-scale land motion and yields a sea-level rate expressed relative to the stable GOM coastal plain.

We analyzed 26 tide-gauge records distributed along the GOM coastline (Fig. 10), drawing RSLRs and their uncertainties (95% CI) from National Oceanic and Atmospheric Administration's Center for Operational Oceanographic Products and Services (NOAA CO-OPS). For each tide gauge, we paired a nearby continuous GNSS station. GNSS data are processed with PPP in the IGS20 frame and aligned to GOM25 to estimate vertical land motion ($\mathrm{VLM}_{\mathrm{GOM25}}$) at the tide-gauge site. Pairing prioritized proximity, record length and continuity, and overall data quality. Displacement time series for the GNSS stations were individually processed using the method of Wang et al. (2022) to carefully remove outliers and correct for instant steps, and the resulting velocities were cross-checked against neighboring stations and published studies (e.g., Hammond et al. 2021) on a case-by-case basis. Across the GOM coast, the mean $\mathrm{SLR}_{\mathrm{GOM25}}$ was 2.72 mm/year for the last half-century (approximately 1970s to 2020s). The standard deviation was ± 0.40 mm/year, representing the spatial variability of local rates along the coastline. The detailed information of the tide-gauge and GNSS pairs along the GOM coast is listed in Table 2.

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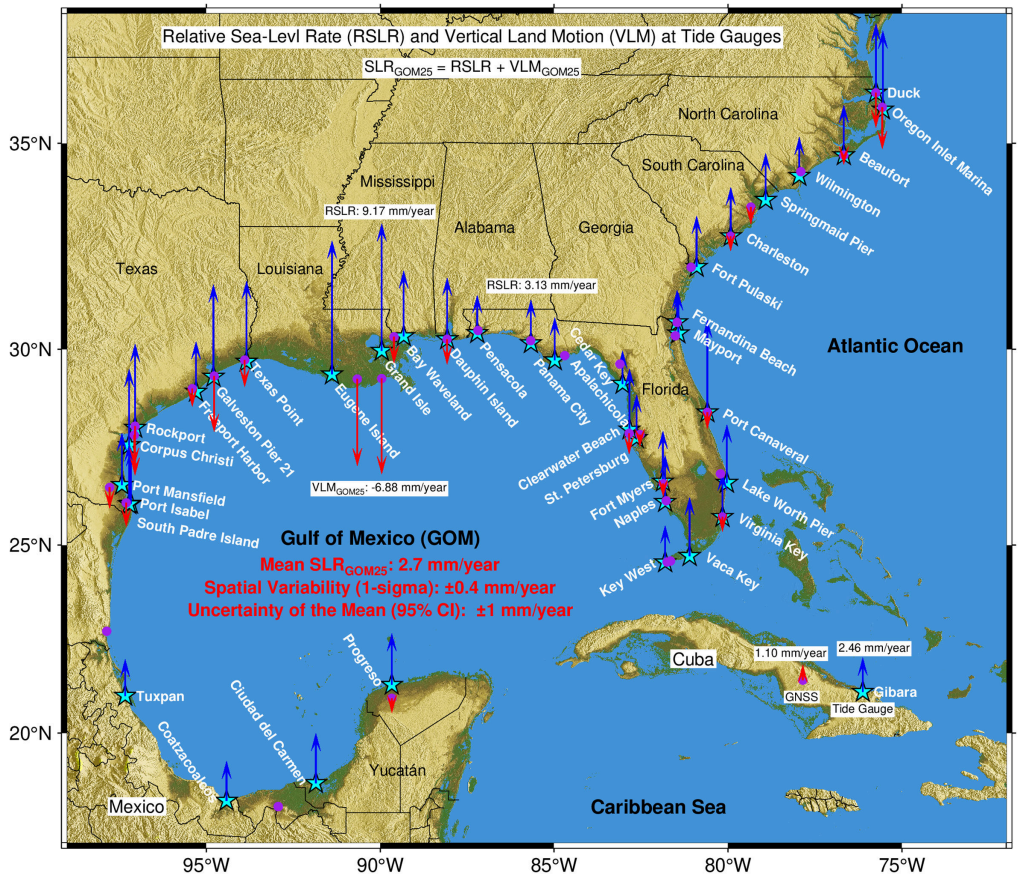


Fig. 10. RSLR and vertical land-motion trends with respect to GOM25 (VLM$_{GOM25}$) estimated from closely spaced tide-gauge/GNSS pairs along the Gulf Coast and the South Atlantic Bight (North Carolina–Florida Atlantic coast). [Topo-bathymetric data from GEOBCO Compilation Group 2024; base map generated using The Generic Mapping Tools, v 6.3.0 (Wessel et al. 2019).]

The uncertainty in each site's sea-level change rate is a function of the record length for both the tide gauge and GNSS data. As empirically modeled in Fig. 3(b), the 95% CI for the SLR uncertainty decays with longer time series. To ensure precision, our analysis is restricted to tide gauge records exceeding 30 years and GNSS segments (change-point free) longer than 4 years. To achieve a 1 mm/year 95% CI for the sea-level rate estimates, approximately 60 years of continuous observations are typically needed (Wang 2023b) [Fig. 3(b)]. Given that the average tide gauge record lengths are approximately 64 years and over 10 years for the GNSS stations (Table 2), the overall uncertainty of the mean SLR$_{GOM25}$ estimate is expected to be at a submillimeter per year level.

To determine if SLR$_{GOM25}$ is consistent across the southeastern US coastline, we extended our analysis to the South Atlantic Bight from North Carolina to Florida coast. Thirteen tide gauge-GNSS pairs along the Atlantic coast were investigated (Fig. 10). The sea-level rate relative to GOM25 at each site is calculated as SLR$_{GOM25}$ = RSLR + VLM$_{GOM25}$. Precise RSLR and VLM$_{GOM25}$ values at each station are given in Tables 2 and 3. Detailed information for these tide gauge and GNSS stations is listed in Table 3. SLR$_{GOM25}$ values at 11 sites predominantly fell within the range of 2.1 to 3.5 mm/year, resulting in a mean rate of 2.73 ± 0.45 mm/year, comparing well with the mean SLR along the GOM coast. This consistency underscores the regional coherence of absolute sea-level rise across the northern Gulf and southeastern Atlantic coasts.

Two notable exceptions to this pattern were identified along the Atlantic coast. Port Canaveral, Florida, exhibited an anomalously high RSLR of 6.65 mm/year. GNSS data close to this site (CCV5 and others) showed a minor subsidence of about 1.0 mm/year, indicating the high trend of the tide-gauge data is likely an artifact of the tide gauge's shorter (~30 years, 1994–2024) record length. Lake Worth Pier, Florida, exhibited a high relative sea-level trend of 4.19 mm/year. The closely spaced GNSS PBCH and others showed high vertical stability (approximately 0.2 mm/year) in this area, resulting in an unexpectedly larger SLR$_{GOM25}$. The tide-gauge data set spans about 50 years (1970–2024) but contains two large data gaps—from 1978 to 1984 and from 1992 to 2009. These discontinuities increase the uncertainty in the estimated RSLR for this site.

In summary, the integration of tide gauge and GNSS data indicates that SLR$_{GOM25}$ along the US GOM coast was approximately 2.7 mm/year during the last half century. The US Southeast Atlantic coast (from North Carolina to Florida) shares a statistically indistinguishable SLR. This critical metric isolates the oceanic component of sea-level change from the confounding signal of localized VLM, providing a clear and precise benchmark essential for long-term sea-level rise assessment, ecosystem management, coastal resilience planning, and infrastructure development onshore and offshore.

Discussion and Conclusions

Role of GOM25 as Geodetic Firmware

A complete geodetic infrastructure comprises three elements: the physical hardware of GNSS stations, the software for data processing

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Table 2. Tide gauge and GNSS pairs for estimating SLR along the Gulf of Mexico coast with respect to GOM25

Tide gauge stations								GNSS stations								
	Location (degrees)		History (years)			RSLR[a] (mm/year)			Location (degrees)		History (years)					
Name	Longitude	Latitude	Start	Stop	Span	Trend	95% CI	Name	Longitude	Latitude	Start	Stop	Span	Distance (km)	GOM25-VLM[b] (m m/year)	GOM25-SLR[c] (m m/year)
Key West, Florida	−81.808	24.551	1913	2024	111	2.64	0.15	FLKW	−81.754	24.554	2002	2025	23	4.8	−0.74	1.90
Vaca Key, Florida	−81.107	24.711	1971	2024	53	4.23	0.40	KYW5	−81.653	24.582	1996	2016	20	58.3	−0.48	3.75
Naples, Florida	−81.808	26.132	1965	2023	58	3.35	0.42	NAPL	−81.776	26.149	2003	2025	22	3.7	−0.90	2.45
Fort Myers, Florida	−81.871	26.648	1965	2024	59	3.64	0.42	FMYR	−81.864	26.591	2004	2025	21	6.4	−0.91	2.73
St. Petersburg, Florida	−82.627	27.761	1947	2024	77	3.13	0.23	MCD6	−82.533	27.850	2001	2019	18	13.1	−0.87	2.26
Clearwater Beach, Florida	−82.832	27.978	1973	2024	51	4.38	0.50	FLIB	−82.843	27.880	2017	2025	8	11.0	−1.49	2.94
Cedar Key, Florida	−83.032	29.135	1914	2024	110	2.38	0.18	XCTY	−83.108	29.631	2004	2025	21	55.6	0.40	2.82
Apalachicola, Florida	−84.982	29.727	1967	2024	57	3.08	0.55	FLCB	−84.695	29.843	2015	2025	10	31.4	−0.20	2.88
Panama City, Florida	−85.667	30.152	1973	2024	51	3.13	0.53	PNMA	−85.673	30.214	2017	2025	8	6.9	−0.10	3.03
Pensacola, Florida	−87.211	30.404	1923	2024	101	2.72	0.23	PCLA	−87.189	30.469	2004	2024	20	7.4	0.12	2.84
Dauphin Island, Alabama	−88.075	30.250	1966	2024	58	4.43	0.53	ALDI	−88.078	30.249	2009	2025	16	0.3	−1.80	2.63
Bay Waveland, Mississippi	−89.325	30.325	1978	2024	46	4.72	0.65	MSIN	−89.604	30.312	2012	2025	13	30.8	−1.90	2.82
Grand Isle, Lousiana	−89.957	29.263	1947	2024	77	9.17	0.36	GRIS	−89.957	29.266	2005	2025	20	0.3	−6.88	2.29
Eugene Island, Louisiana	−91.385	29.372	1939	1974	35	9.65	1.24	LMCN	−90.661	29.255	2003	2025	22	80.9	−6.26	3.39
Texas Point, Texas	−93.842	29.690	1958	2024	66	5.81	0.62	TXSP	−93.897	29.731	2016	2025	9	6.3	−2.90	2.91
Galveston Pier 21, Texas	−94.793	29.310	1904	2024	120	6.65	0.21	TXGA	−94.773	29.328	2005	2025	20	2.8	−4.00	2.65
Freeport Harbor, Texas	−95.295	28.935	1972	2024	52	3.62	0.6	DWI1	−95.404	29.014	2009	2025	16	12.3	−1.28	2.34
Rockport, Texas	−97.047	28.022	1937	2024	87	6.05	0.44	TXRP	−97.049	28.062	2013	2017	4	4.5	−3.53	2.52
Corpus Christi, Texas	−97.217	27.580	1983	2022	39	5.48	0.98	TXPO	−97.070	27.839	2010	2025	15	31.4	−2.21	3.27
Port Mansfield, Texas	−97.426	26.558	1963	2024	61	3.75	0.61	TXRV	−97.781	26.494	2013	2025	12	38.0	−1.50	2.25
South Padre Island, Texas	−97.168	26.073	1958	2018	60	4.00	0.69	TXLN	−97.301	26.095	2013	2021	8	14.6	−1.70	2.30
Port Isabel, Texas	−97.216	26.061	1944	2024	80	4.37	0.31	TXLN	−97.301	26.095	2013	2021	8	9.6	−1.70	2.67
Tuxpan, Mexico	−97.333	21.000	1958	1990	32	2.64	1.45	TAM1	−97.864	22.278	2009	2022	13	150.7	0.16	2.80
Coatzacoalcos, Mexico	−94.417	18.150	1952	1987	35	2.86	1.06	VIL2	−92.931	17.990	2010	2021	11	165.0	−0.45	2.41
Ciudad del Carmen, Mexico	−91.850	18.633	1956	1988	32	3.60	0.94	VIL2	−92.931	17.990	2010	2021	11	132.0	−0.45	3.15
Progreso, Mexico	−89.667	21.300	1952	2013	61	3.69	0.78	MMD1	−89.663	20.932	2008	2025	17	41.0	−1.04	2.65
Average					64	—	0.58	—	—	—	—	—	15	—	—	2.72 ± 0.40

[a]RSLR: Relative sea-level rate (mm/year) with 95% confidence interval (95% CI), sourced from NOAA/CO-OPS. Data from 26 tide gauges along the GOM coast.
[b]GOM25-VLM: Vertical land motion rate (mm/year) with respect to GOM25. The uncertainties (95% CI) are below 1 mm/year.
[c]GOM25-SLR: Sea-level rate with respect to GOM25, the stable Gulf Coastal Plain.

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Table 3. Tide gauge and GNSS Pairs for estimating sea-level rate (SLR) along the South Atlantic Bight (North Carolina to Florida Coast) with respect to GOM25

Tide gauge stations								GNSS stations								
	Location (degrees)		History (years)			RSLR[a] (mm/year)			Location (degrees)		History (year)				GOM25-VLM[b]	GOM25-SLR[c]
Name	Longitude	Latitude	Start	Stop	Span	Trend	95% CI	Name	Longitude	Latitude	Start	Stop	Span	Distance(km)	(m m/year)	(m m/year)
Duck, North Carolina	−75.747	36.183	1978	2024	46	4.99	0.56	NCDU	−75.751	36.182	2008	2024	16	0.1	−2.50	2.49
Oregon Inlet Marina, North Carolina	−75.548	35.795	1977	2024	47	5.67	0.95	NCBI	−75.564	35.846	2013	2021	8	5.7	−3.10	2.57
Beaufort, North Carolina	−76.670	34.717	1953	2024	71	3.62	0.34	NCBE	−76.672	34.719	2008	2025	17	0.3	−0.66	2.96
Wilmington, North Carolina	−77.953	34.227	1935	2024	89	2.79	0.34	NCWG	−77.908	34.329	2016	2025	9	13.5	−0.10	2.69
Springmaid Pier, South Carolina	−78.918	33.655	1957	2024	68	3.36	0.47	KNS6	−79.343	33.482	2003	2019	16	45.0	−1.23	2.13
Charleston, South Carolina	−79.923	32.782	1901	2024	124	3.51	0.19	SCHA	−79.925	32.779	2009	2018	9	0.2	−1.05	2.46
Fort Pulaski, Georgia	−80.903	32.035	1935	2024	90	3.66	0.27	GASA	−81.060	32.024	2015	2025	10	0.1	−0.25	3.41
Fernandina Beach, Florida	−81.467	30.672	1897	2024	128	2.31	0.17	FLFB	−81.466	30.668	2017	2025	8	0.1	0.24	2.55
Mayport, Florida	−81.428	30.398	1928	2024	97	2.93	0.24	KREG	−81.515	30.336	2017	2025	8	13.6	0.20	3.13
Virginia Key, Florida	−80.162	25.732	1931	2024	94	3.19	0.22	N300	−80.162	25.731	2018	2025	7	0.1	−1.05	2.14
Gibara, Cuba	−76.125	21.108	1974	2021	47	2.46	0.74	CN16	−77.850	21.422	2014	2024	10	190.0	1.10	3.46
Average:					82	3.50	0.41	—	—	—	—	—	11	—	−0.76	2.73 ± 0.45
Port Canaveral, Florida	−80.593	28.415	1994	2024	31	6.65	1.34	CCV5	−80.545	28.460	1998	2019	21	7.0	−1.05	5.60
Lake Worth Pier, Florida	−80.033	26.613	1970	2024	55	4.19	0.50	PBCH	−80.219	26.846	2005	2025	20	26	0.20	4.39

[a]RSLR: relative sea-level rate with 95% confidence interval (95% CI), provided by NOAA/CO-OPS. Data from 13 tide gauges along the South Atlantic coast.
[b]GOM25-LVM: Land vertical movement rate (subsidence) with respect to GOM25. The uncertainties (95% CI) are below 1 mm/year.
[c]GOM25-SLR: Sea-level rate with respect to GOM25, the stable Gulf Coastal Plain.

(e.g., PPP, calculating site velocities and uncertainties), and the fundamental reference frame, or firmware. GOM25 serves as this critical firmware, a stable, long-term foundation that remains consistent for years while maintaining alignment with global standards. Its exceptional stability, approximately 0.3 mm/year horizontally and 0.5 mm/year vertically, ensures precise and reliable ground and structural deformation monitoring in the region. The seamless transformation of ECEF-XYZ, ENU displacement time series, and site velocities between IGS20 and GOM25 via the one-set seven-parameter transformation further enhances its utility and accessibility.

Unifying Platform for Multidisciplinary Integration

GOM25 provides a unified platform for integrating diverse geodetic and remote-sensing observations into a consistent spatial context, which is pivotal for both scientific research and operational applications. The extensive GNSS time series within the GOM25 frame serves as an invaluable resource, enabling hydrologists to quantify ground deformation from aquifer compaction, geomorphologists to investigate coastal erosion and wetland dynamics, and oceanographers to calibrate long-term sea-level change measurements. Furthermore, GOM25-derived absolute displacements provide essential ground truth for calibrating and validating relative measurements from techniques like InSAR, LiDAR, and UAV photogrammetry. This synergy, particularly the powerful combination of GNSS and InSAR, significantly enhances the reliability of subsidence assessments and constrains spatially extensive deformation fields across the region.

By delivering a stable, unified datum, GOM25 empowers authorities to base decisions on consistent, high-precision data that are comparable across different states and jurisdictions. This capability enables robust cross-region analysis of spatial patterns—such as differential subsidence rates threatening pipelines, transportation networks, and water supply infrastructure—and supports long-term temporal monitoring by eliminating the need for costly local reference networks. This makes high-precision deformation monitoring accessible for safeguarding critical infrastructure, including assessing the stability of energy pipelines and ports, monitoring the health of bridges and levees over decades, regulating development in subsidence-prone areas with consistent standards, and managing groundwater extraction to protect against foundation damage.

Extensive Spatial Applicability and Validation

This study establishes that GOM25 provides a physically valid reference across an extensive geographical domain. Although the Helmert transformation can be mathematically applied anywhere, the physical validity of GOM25 for deformation studies is defined by the region where a locally stable site exhibits near-zero velocities (e.g., <1 mm/year) in all three directions. Euler pole analysis confirms the transformation successfully removes the large-scale rigid plate motion. We confidently define this applicable domain from 15° to 37° N and 110° to 75° W, the geographic extent depicted in Fig. 7. The stability and utility of the frame have been rigorously validated across this entire region.

Crucially, validation also confirmed that GOM25 performs robustly not only within the core network on the US portion of the GOM coast but also across the stable southern North American Plate, including the GOM basin, the Yucatán Platform, and Cuba.

Summary and Outlook

GOM25's value lies in its dual role: it provides a spatially consistent framework for understanding regional-scale processes and a temporally stable benchmark for tracking long-term ground deformation. By enabling direct comparison of ground deformation rates across the region—from New Mexico to North Carolina and Mexico to Cuba—GOM25 integrates isolated local studies into a coherent regional assessment. This enables environmental and infrastructure decisions across the GOM region to be based on a unified and stable geodetic standard. The improved accuracy, stability, and spatial extent of GOM25, combined with its enhanced accessibility through the permanently archived open-source Python tools and data sets (Wang and Welch 2026), make it an indispensable platform for interdisciplinary geoscience research, coastal hazard mitigation, infrastructure health monitoring, and long-term regional planning. As future IGS reference frames are released, GOM25 will be incrementally updated to maintain consistency with the prevailing global reference frame, preserving its utility for the region.

Data Availability Statement

The GNSS position time series (ECEF-XYZ) in the IGS20 reference frame used in this study are publicly available from the Nevada Geodetic Laboratory (NGL) at http://geodesy.unr.edu. All products generated in this study—including the complete list of 150 reference stations, Python transformation tools, and ENU displacement time series for approximately 1,400 GNSS stations in both IGS20 and GOM25—are permanently archived in Wang and Welch (2026). The archive includes primary Python tools for coordinate transformation, displacement and velocity conversion, and Euler pole calculation, along with accompanying documentation and time series plots.

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Author Contributions

Jennifer Welch: Data curation; Formal analysis; Investigation; Methodology; Writing – review and editing. Guoquan Wang: Conceptualization; Data curation; Formal analysis; Methodology; Writing – original draft.

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